

MSc Applied Artificial Intelligence
Dissertation Project Implementation Plan

WMG, University of Warwick

A Structured Guide for Navigating the Research and Writing Process

Contents

Preamble: How to Use This Document	2
Summary: An estimation of word count allocations	2
Abstract.....	3
Checklist.....	3
Common Pitfalls	3
Chapter 1: Introduction.....	4
Checklist.....	4
Common Pitfalls	5
Chapter 2: Literature Review.....	5
Checklist.....	5
Structure and Coverage	5
Criticality and Depth.....	6
References and Citations.....	6
Common Pitfalls	6
Chapter 3: Methodology	6
Checklist.....	7
Research Design	7
Data	8
Model and Algorithm Selection.....	8
Training and Optimisation	8
Evaluation Design	8
Common Pitfalls	8
Chapter 4: Analysis and Results	9
Checklist.....	10
Presentation of Results	10
Evaluation Rigour.....	10
Integrity.....	10
Common Pitfalls	11
Chapter 5: Discussion	11
Checklist.....	11
Interpretation.....	11

Limitations.....	12
Implications.....	12
Future Work	12
Common Pitfalls	12
Chapter 6: Conclusion.....	12
Checklist.....	13
Common Pitfalls	13
Reference List: Standards and Pitfalls	14
Standards to Meet	14
Common Reference List Pitfalls.....	14
Final Reflections for Students	14

Preamble: How to Use This Document

This implementation plan is intended as your ongoing reference throughout the dissertation process. It is not a template to be followed mechanically, but a framework that helps you self-audit your work at each stage. Return to the checklists repeatedly at the drafting stage, after supervisor feedback, and before final submission.

The quality of an MSc dissertation in Applied AI is determined not just by the sophistication of your model or system, but by the rigour of your reasoning, the coherence of your narrative, and the honesty of your evaluation. Each chapter has a distinct role, and the most common failing is treating them as isolated documents rather than chapters of a single, unified argument.

Summary: An estimation of word count allocations

Chapter	Suggested Word Count
Abstract	300–350 (not in 15,000 total)
1. Introduction	1,000–1,200
2. Literature Review	3,000–3,500
3. Methodology	2,500–3,000
4. Analysis and Results	2,000–2,500
5. Discussion	2,500–3,000
6. Conclusion	800–1,000
Total (approximate)	~13,800–14,200 (max: 15,000)

Do not treat word counts as targets to pad toward. A tightly argued 13,500-word dissertation will always outscore a meandering 15,200-word one.

Abstract

The abstract is a self-contained summary of the entire dissertation. It should be written last, even though it appears first. Its purpose is to allow a reader to determine whether your dissertation is relevant to their interests and what you found. Think of it as a highly compressed version of your research story: problem, approach, findings, and significance. In Applied AI dissertations, the abstract must convey not just what you built, but how it performed and what that performance means in context. This section must be written as a single paragraph, and include keywords at the end of it.

Suggested Word Count

300–350 words — not included in the 15,000-word total

Checklist

- ☐ States the research problem or question clearly in the first two sentences.
- ☐ Identifies the gap in knowledge or practice that the dissertation addresses.
- ☐ Names the dataset(s), domain, and the AI approach or technique used.
- ☐ Summarises the methodology in one or two sentences without excessive technical jargon.
- ☐ Reports at least one key quantitative finding (e.g., accuracy, F1-score, BLEU score, RMSE) with enough context to be interpretable.
- ☐ States what the findings mean — not just what was achieved, but why it matters.
- ☐ Does not contain citations, acronyms that are not spelt out, or undefined abbreviations.
- ☐ Is written in the past tense for completed work and present tense for conclusions that remain true.
- ☐ Is between 300 and 350 words.
- ☐ Can be read and understood independently of the rest of the dissertation.

Common Pitfalls

- Writing the abstract too early and failing to update it to reflect actual findings.
- Writing as a vague statement of intentions ('this dissertation will investigate...') rather than a factual account of what was done and discovered.
- Omitting any numerical results, which leaves the abstract with no evidential weight.

Chapter 1: Introduction

The introduction is your opportunity to frame the research problem within a broader context and to persuade the reader that your dissertation is worth reading. It is not a literature review, and it is not the place to explain technical methods in detail. Its job is to establish the why before the how.

For this purpose, you must clarify the context in which you project the scope. This means situating your project within a real-world domain and articulating a specific problem that AI methods are positioned to address. The introduction concludes by stating your research questions or objectives clearly, giving the reader a navigational map for everything that follows.

Suggested Word Count

1,000–1,200 words

Checklist

- ☐ Opens with a compelling and factually grounded contextualisation of the problem domain. Avoid opening with generic statements such as 'Artificial intelligence is transforming every industry'. It must contextualise the problem you will solve.
- ☐ Establishes the practical or societal significance of the problem being investigated.
- ☐ Narrows progressively from a broad domain context to the specific research problem (funnel structure).
- ☐ Identifies a clear gap, limitation, or unresolved challenge in existing practice or research. It is not necessary to innovate in the master's, but you need at least a clear contribution.
- ☐ States research aims and objectives in precise, actionable language. Usually, you define one main objective, related to your research questions, and different tasks to achieve this main objective.
- ☐ States research questions explicitly, or alternatively frames the problem in terms of testable hypotheses.
- ☐ Briefly describes the methodology and approach at a high level — one short paragraph is sufficient.
- ☐ Briefly describes the dataset or application context without going into methodological detail.
- ☐ Provides a chapter-by-chapter outline in the final paragraph.
- ☐ Every claim made about the problem domain is supported by a citation to a credible source.
- ☐ Does not confuse the introduction with a literature review.
- ☐ Does not contain code, figures of model architectures, or tables of results.

Common Pitfalls

- Stating research objectives so vaguely that they cannot be evaluated at the end of the dissertation. Objectives such as 'to explore machine learning' are topics, not objectives.
- Conflating motivation with literature review by summarising entire bodies of prior work in the introduction.
- Failing to align objectives in the introduction with what is actually evaluated in the Analysis and Conclusion chapters.

Chapter 2: Literature Review

The literature review demonstrates that you have engaged critically with the existing body of knowledge relevant to your project. It is emphatically not a series of summaries of individual papers. A high-quality literature review synthesises evidence and perspectives across multiple sources, identifies agreements and contradictions, traces how understanding of a topic has evolved, and ultimately builds a compelling case for why your specific research approach is justified.

In Applied AI dissertations, the literature review typically covers: the theoretical foundations of the AI techniques being used, prior applications of those techniques in your domain of interest, the limitations and gaps in existing work, and — critically — the criteria and evaluation metrics that prior work has used to assess performance.

At the end of it, you must write a paragraph that summarizes how your contribution and chosen methodology links to the literature.

Suggested Word Count

3,000–3,500 words

Checklist

Structure and Coverage

- ☐ Organised thematically or conceptually, not as a sequential list of 'Paper A says... Paper B says...'
- ☐ Covers the theoretical underpinnings of the AI methods used (e.g., transformer architectures, reinforcement learning theory, Bayesian inference).
- ☐ Reviews prior applications of similar AI methods in the same or closely related domains.
- ☐ Addresses evaluation benchmarks and performance metrics used in comparable studies.

- ☐ Discusses ethical, societal, or deployment considerations where relevant to the project.
- ☐ Concludes with a synthesis section that clearly articulates the gap your research addresses.

Criticality and Depth

- ☐ Evaluates rather than merely describes — identifies strengths, weaknesses, and limitations of prior work.
- ☐ Draws comparisons across studies: what do they collectively suggest or leave unresolved?
- ☐ Avoids over-reliance on a single source or a narrow set of authors.
- ☐ Prioritises recent literature (last 3 years) as the primary base, supplemented by foundational older works where appropriate.
- ☐ Does not treat textbook content or Wikipedia as primary academic sources.
- ☐ Does not rely heavily on grey literature (blog posts, vendor documentation) without acknowledging limitations.

References and Citations

- ☐ Uses a minimum of 10–15 peer-reviewed sources for a dissertation of this scope.
- ☐ Cites primary sources rather than citing papers that cite other papers.
- ☐ All citations are formatted consistently in the required referencing style (typically Harvard).
- ☐ Every claim that is not common knowledge is supported by a citation.
- ☐ Page numbers are included where required (e.g., for direct quotations in Harvard style).

Common Pitfalls

- Annotation rather than synthesis — producing what is effectively a list of abstracts rather than a coherent analytical argument.
- Recency imbalance: relying almost entirely on papers from several years ago in a fast-moving field.
- Secondary citations — citing a paper's description of another paper's findings without reading the original.
- A literature review that makes no connection to the methodology, failing to justify technical choices.

Chapter 3: Methodology

The methodology chapter explains how you conducted your research. It must be detailed enough that a competent peer in your field could replicate what you did, but it must also justify each decision rather than merely describing it. This distinction between description and justification is essential: every choice — of dataset, algorithm, architecture, hyperparameter strategy, evaluation metric — should be accompanied by a rationale grounded in your literature review or in principled reasoning.

How Students Typically Write This Chapter And Why That Is a Problem

The most common mistake is writing a methodology that reads as a technical report or a code walkthrough. Students frequently dedicate large portions of this chapter to reproducing code snippets, explaining what a random forest is (which belongs in the literature review), or listing every library used.

What the methodology must instead focus on is why you made the choices you made and how those choices constitute a valid approach to your research question.

A second common problem is treating the methodology as a diary of what happened during the project. Methodology is structured around logical, defensible decision-making, not chronology.

A third issue is the failure to describe the research design at an appropriate level of abstraction before diving into technical details. Students should briefly position their work within a research paradigm (e.g., design science research, experimental comparison, case study) before discussing model selection and training.

Suggested Word Count

2,500–3,000 words

Checklist

Research Design

- ☐ Identifies and justifies the overall research design or paradigm (e.g., experimental, design science, comparative evaluation).
- ☐ Explains whether the work is predictive, generative, classification-based, optimisation-based, or otherwise, and why this framing suits the research question.
- ☐ Describes ethical considerations — consent, data privacy, bias, and fairness — where applicable.

Data

- ☐ Describes the dataset(s) fully: source, size, format, collection method, time period, and any known limitations or biases.
- ☐ Explains how data was split (train/validation/test) and justifies the split strategy (e.g., stratified k-fold, time-based split for temporal data). Note, you explain only the method (e.g., for splitting data) rather than adding results of your implementation.
- ☐ Describes all preprocessing and feature engineering steps with justifications, not just descriptions. Think about this as writing a guidance for others to replicate your work, considering they will use a different dataset.
- ☐ Addresses class imbalance, missing values, or data quality issues and explains how these were handled.

Model and Algorithm Selection

- ☐ Justifies the choice of model(s) or algorithms with reference to the literature review.
- ☐ Explains why alternative approaches were not selected (comparative justification, not dismissal).
- ☐ Describes baseline models used for comparison and explains why they were chosen as baselines.
- ☐ Describes the architecture or configuration of the model(s) in sufficient detail for replication.

Training and Optimisation

- ☐ Explains the hyperparameter tuning strategy (e.g., grid search, random search, Bayesian optimisation) and justifies the choice.
- ☐ Describes regularisation strategies and why they were appropriate.
- ☐ Notes hardware/computational environment if relevant to reproducibility.
- ☐ Describes how overfitting was monitored and mitigated.

Evaluation Design

- ☐ Names all evaluation metrics used and justifies each with reference to the research question and the literature.
- ☐ Explains the evaluation protocol (e.g., held-out test set, cross-validation, statistical significance testing).
- ☐ Addresses the risk of data leakage and explains how it was prevented.

Common Pitfalls

- Absent justification: describing what was done but not why, leaving the examiner unable to assess whether choices were principled or arbitrary.
- Treating deep learning architectures or pre-trained models as black boxes that are simply 'used' without explaining why they are appropriate.
- Students using transfer learning or large pre-trained models (GPT variants, BERT, ResNet) must explain fine-tuning and domain transferability assumptions.
- Reporting results in the methodology chapter — results belong exclusively in the Analysis and Results chapter.

Chapter 4: Analysis and Results

This chapter presents what you found, with objectivity, precision, and appropriate structure. Its purpose is to report the outcomes of applying your methodology to your data, not to interpret or contextualise those outcomes in depth — that work belongs in the Discussion. The distinction matters: results are facts; discussion is meaning. This chapter validates your approach and evaluates your results against your main objective and research questions.

How Applied AI Projects Are Validated and Evaluated

Supervised learning: Report accuracy, precision, recall, F1-score, AUC-ROC, RMSE, MAE on a held-out test set, with confusion matrices, ROC curves, or calibration plots.

Comparative experiments: Present results across models in tables, ideally with statistical significance tests (e.g., McNemar's test, paired t-tests) to confirm differences are not due to chance.

Generative AI / NLP: Combine automatic metrics (BLEU, ROUGE, BERTScore, perplexity) with human evaluation components.

Reinforcement learning: Use learning curves, reward convergence plots, and comparison of the trained agent against a baseline or random agent.

Anomaly detection / unsupervised systems: Rely on domain-expert validation, precision-recall at various thresholds, or established benchmarks.

In all cases, a single metric is almost never sufficient. Ablation studies — removing or replacing individual system components to quantify their contribution — are strongly encouraged and significantly elevate analysis quality.

Suggested Word Count

2,000–2,500 words

Checklist

Presentation of Results

- ☐ Results are presented in a logical order that mirrors the structure of the methodology chapter.
- ☐ All tables and figures are numbered, clearly titled, and accompanied by a descriptive caption.
- ☐ Tables and figures are referenced and discussed in the text — never presented without commentary.
- ☐ Quantitative results are reported with appropriate precision (e.g., to two decimal places for percentages).
- ☐ Confidence intervals or standard deviations are reported where applicable, especially for cross-validated results.

Evaluation Rigour

- ☐ Performance is reported on a properly held-out test set, not on the validation set used during tuning.
- ☐ Results for the proposed model are compared against at least one baseline.
- ☐ Multiple evaluation metrics are reported, including the primary metric justified in the methodology.
- ☐ Statistical significance of key comparisons is addressed where sample sizes permit.
- ☐ Ablation studies or component-level analysis are included to demonstrate the contribution of individual design choices.
- ☐ Learning curves, confusion matrices, or equivalent diagnostic visualisations are provided where appropriate.
- ☐ Failure cases, misclassifications, or anomalous results are reported honestly rather than omitted.

Integrity

- ☐ No selective reporting of results — no omitting of metrics that do not support your hypothesis.
- ☐ Discrepancies between expected and actual results are acknowledged.
- ☐ Limitations of the evaluation (e.g., small test set, single dataset) are briefly acknowledged.

Common Pitfalls

- Cherry-picking results — reporting only favourable metrics. This is not merely a stylistic problem; it is a research integrity issue.
- Evaluating on the validation set after hyperparameter tuning, which produces optimistically biased estimates of generalisation performance.
- Failing to compare results with prior work from the literature review, missing the opportunity to contextualise whether performance is competitive or state-of-the-art. This is not essential, unless you want to publish a paper.
- Results sections consisting almost entirely of figures and tables with minimal analytical narration.

Chapter 5: Discussion

The discussion is the intellectual heart of the dissertation. It is where you step back from the specifics of your results and ask: what do these findings actually mean? This involves interpreting your results in light of your research questions, comparing your findings with prior literature, explaining unexpected outcomes, and acknowledging the limitations of your work with candour.

In Applied AI dissertations, the discussion should also address the practical implications of your findings — what would it mean to deploy this system in the real world? What are the ethical dimensions? The discussion is your opportunity to demonstrate the kind of critical, reflective thinking that distinguishes an MSc-level dissertation from a technical project report.

Suggested Word Count

2,500–3,000 words

Checklist

Interpretation

- ☐ Directly addresses each research question or objective stated in the introduction and provides a substantiated answer.
- ☐ Interprets results rather than restating them — explains what the findings mean, not what the numbers are.
- ☐ Compares findings explicitly with prior work cited in the literature review (confirming, contradicting, or extending prior findings).
- ☐ Explains unexpected or counterintuitive results with reference to evidence or plausible mechanisms.
- ☐ Avoids overclaiming — findings are presented with appropriate epistemic humility and qualified language.

Limitations

- ☐ Discusses the limitations of the dataset (size, representativeness, temporal scope, potential biases).
- ☐ Discusses the limitations of the methodology (assumptions made, generalisability of the approach).
- ☐ Discusses the limitations of the evaluation (e.g., absence of real-world deployment testing, narrow benchmark scope).
- ☐ Limitations are presented analytically — not as an apology, but as an honest assessment that guides interpretation.

Implications

- ☐ Discusses the practical implications of the findings for the relevant domain or industry.
- ☐ Addresses ethical considerations: fairness, bias, transparency, accountability, and data privacy where applicable.
- ☐ Discusses what would be required to deploy or operationalise the system beyond the research context.

Future Work

- ☐ Identifies specific, actionable directions for future research grounded in the limitations and findings.
- ☐ Avoids vague suggestions like 'future work could use more data' — proposals should be precise and justified.

Common Pitfalls

- Repetition of results without interpretation — restating what the confusion matrix showed rather than explaining what it means for the domain.
- Absence of connection to the literature: if prior work achieved a certain benchmark and your results differ significantly, the discussion must engage with that gap.
- Treating limitations superficially as a formulaic gesture rather than engaging with what they imply for the validity and generalisability of conclusions.
- Under-engaging with questions of model fairness and societal impact — examiners increasingly expect substantive ethical reflection, not a single-sentence acknowledgement.

Chapter 6: Conclusion

The conclusion brings the dissertation to a close by synthesising everything that has been established. It is not a summary of the literature review or the methodology. Its

role is to consolidate what was learned, to confirm whether the research aims and objectives were achieved, and to leave the reader with a clear sense of the dissertation's contribution to knowledge or practice.

In an Applied AI context, the conclusion should clearly state what the project demonstrated about the suitability, performance, or limitations of the AI approach chosen, and what this means for the field or domain.

Suggested Word Count

800–1,000 words

Checklist

- ☐ Opens with a brief restatement of the research problem and the motivation for the project.
- ☐ Revisits each research objective or question and states explicitly — with evidence — whether and to what extent it was achieved.
- ☐ Summarises the principal findings without introducing new information or new citations.
- ☐ States the dissertation's contribution clearly: what does the reader now know that they did not know before?
- ☐ Acknowledges the most significant limitations in a way that contextualises rather than undermines the contribution.
- ☐ Proposes two or three specific, well-reasoned directions for future work.
- ☐ Ends with a closing statement that reflects on the broader significance of the work.
- ☐ Does not introduce new arguments, new evidence, or new references.
- ☐ Does not begin with 'In conclusion...' — this is redundant when the chapter heading already signals the conclusion.

Common Pitfalls

- Conclusions that are too brief — a single paragraph is not sufficient at MSc level.
- Introducing new ideas or cited work in the conclusion — it must work entirely from what has already been established.
- Failing to return explicitly to all research objectives stated in the introduction.

Reference List: Standards and Pitfalls

The reference list carries no word count, but its quality significantly influences the examiner's overall impression of the dissertation.

Standards to Meet

The reference list must be formatted entirely consistently in your required style (Harvard). Every source cited in the body must appear in the reference list, and every entry in the reference list must correspond to an in-text citation. Aim for a minimum of 20–30 references for a dissertation of this scope, with the majority drawn from peer-reviewed journals and conference proceedings. Appropriate venues for Applied AI work include NeurIPS, ICML, ICLR, ACL, AAAI, IEEE, and ACM, among others.

Common Reference List Pitfalls

- Inconsistent mixing of referencing styles — some entries formatted in Harvard, others in APA or IEEE.
- Over-reliance on web sources: blog posts, Medium articles, vendor documentation, and GitHub repositories cannot substitute for peer-reviewed literature as the primary evidence base.
- Citing Wikipedia is not acceptable in an MSc dissertation under any circumstances.
- Citing arXiv preprints is acceptable but should be done with awareness that preprints have not been peer-reviewed; this should be acknowledged when the claim is substantive.
- Citing review articles or textbooks as the primary evidence for claims that should be supported by original research papers — this signals shallow engagement with the primary literature.
- Failing to include Digital Object Identifiers (DOIs) or URLs for online sources, making references impossible to verify.
- Secondary citations — citing a paper's description of another paper's findings without reading the original.
- Non-existent or inconsistent sources generated by AI will send you to an ACP.

Final Reflections for Students

The quality that most consistently separates a merit from a distinction in a dissertation is coherence: the degree to which the research question, the literature review, the methodological choices, the evaluation design, the results, and the discussion form a single, logically interconnected argument. Each chapter should feel inevitable given the one that preceded it.

If your methodology chapter introduces a technique that was never discussed in the literature review, or your discussion addresses a research question that was never

stated in the introduction, the dissertation has a structural problem that no amount of technical sophistication will compensate for.

Core Principle

Build your dissertation as an argument, not a collection of deliverables.

Every chapter must connect logically to the one before it and the one after it. The examiner should be able to trace a single coherent intellectual thread from your first research objective to your final conclusion.